

Prakash Sai Alla

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SUMMARY

Business Analytics Graduate Student at UMass Amherst specializing in data analysis, predictive modeling and BI reporting. Built ML models at 84% accuracy, automated ETL pipelines, reducing analyst time by 20% and surfaced revenue insights from over 50,000 records. Skilled in SQL, Python, Power BI and Tableau - delivering dashboards and stakeholder reports.

SKILLS

Technical Skills: Python, Advanced SQL, Excel (Pivot Tables, VLOOKUP), Power BI, Tableau, R, Git, Streamlit
Data & Analytics: Data Visualization, Data Modeling, Predictive Analytics, Statistical Analysis, Machine Learning
Business Intelligence: Dashboard Development, KPI Design, Data Storytelling, Executive Reporting, AI Agents
Data Engineering: ETL Processes, Data Cleaning & Transformation, Data Warehousing, Microsoft Fabric, Delta Lake
Soft Skills: Business Communication, Stakeholder Management, Problem Solving, Critical Thinking, Team Collaboration

EXPERIENCE

Business Analyst

Sep 2024 – May 2026

UMass Auxiliary Enterprises

Amherst, US

- Analyzed KPIs, budgets, financial transactions and participation trends across 10+ campus initiatives, generating actionable insights that contributed to a 20% increase in student engagement and programs serving 500+ students annually.
- Conducted budget tracking, expense analysis and stakeholder coordination across 15+ cross-functional team members and volunteers, ensuring efficient resource allocation and successful execution of 8+ large scale campus events.
- Developed data driven outreach strategies by translating engagement and financial performance data into actionable recommendations, improving event visibility, participant experience and overall program effectiveness.

Machine Learning and Data Science Intern

Apr 2023 – Jul 2023

PHN Technologies Pvt Ltd

Hyderabad, IN

- Audited and transformed 5+ datasets using Python and SQL, improving reporting accuracy by 15% through anomaly detection, data validation, workflow optimization and analytical quality assurance processes.
- Automated ETL pipelines and preprocessing workflows across 5+ datasets, reducing analyst manual effort by 20% while improving operational efficiency, reporting consistency and scalable analytics processing capabilities.
- Built 3 machine learning models using Python, Pandas and Scikit-learn, outperforming baseline performance by 12% while delivering stakeholder focused dashboards, predictive insights and business decision-making recommendations.

PROJECTS

CareerPilot AI

Python, Streamlit, Azure OpenAI, Azure AI Foundry

- Developed and deployed an AI powered career guidance platform for the Microsoft Agents League Hackathon, delivering resume analysis, skill gap assessment, interview preparation and personalized career recommendations.
- Designed a scalable multi agent architecture with Azure AI Foundry and Streamlit, integrating fallback AI support and generating career insights, certification recommendations and learning plans through an interactive user facing application.

Health Insurance Premium Prediction

Python, Scikit-learn, SQL

- Developed a multiple regression model using Scikit-learn and SQL on 10+ demographic and medical risk variables, achieving an R^2 score of 0.84 for health insurance premium prediction and actuarial risk analysis.
- Built an end to end predictive analytics pipeline including data cleaning, feature engineering and model evaluation, identifying Age and BMI as key premium drivers while delivering stakeholder focused insights and visual reports.

Airbnb Demand & Availability Forecasting

Python, Pandas, Time-Series Analysis

- Collected and analyzed 50,000+ Airbnb listings using Python and Pandas, engineering time-series features to identify seasonal demand patterns, pricing trends and occupancy behavior across high demand urban markets.
- Developed demand forecasting models and stakeholder focused analytical reports that identified revenue optimization opportunities, enabling data driven pricing strategies and improved occupancy performance benchmarking.

EDUCATION

Master of Science, Business Analytics

GPA: 4.0

University of Massachusetts Amherst, Isenberg School of Management

Sep 2024 – May 2026

Bachelor of Technology, Artificial Intelligence and Machine Learning

Jawaharlal Nehru Technological University Hyderabad

Nov 2020 – Aug 2024

CERTIFICATIONS

- Microsoft Azure AI Fundamentals (AI-900)
- Career Essentials in Data Analysis – Microsoft / LinkedIn
- Machine Learning & Data Analytics– Infosys
- Silver Medal in NPTEL's The Joy of Computing with Python – IIT Madras